

0.1 Standard Model of Particle Physics

The Standard Model (SM) of particle physics is a framework which contains our current understanding of the matter and three of the four forces found in the universe. Within the SM there are 12 elementary fermions, 5 vector bosons which mediate the three fundamental forces and a scalar boson responsible for particle masses (Figure 1). It is important to note that the SM is not a complete model of the universe. A great example of this is the gravitational force which is not described by the theory, however at the small scales of particle physics this force is neglected due to its insignificant strength when compared to the other three forces (strong, electromagnetic and weak).

Quarks	u Up	c Charm	t Top	g Gluon	H Higgs
	d Down	s Strange	b Bottom	γ Photon	Gauge Bosons
Leptons	e Electron	μ Muon	τ Tau	Z Z Boson	
	ν_e Electron Neutrino	ν_μ Muon Neutrino	ν_τ Tau Neutrino	W W Boson	

Figure 1: The Standard Model of particle physics with the 12 fermions shown in purple and green, the five vector bosons shown in red and the Higgs scalar boson shown in yellow.

The SM comprises of a quantum field theory (QFT) and is based on the non-Abelian gauge symmetry groups $SU(3) \times SU(2) \times U(1)$. The $SU(3)$ governs the strong interactions and the $SU(2) \times U(1)$ the electroweak interactions. Each group contains their respective gauge bosons: gluon for the strong and W, Z and photon for the electroweak. The $SU(3)$ is an exact gauge symmetry where as the electroweak symmetry is spontaneously broken by the Higgs field, resulting in the massive electroweak bosons ($W^\pm$ and $Z^\pm$). Due to this broken symmetry which gives rise to massive

quarks, the strong and weak eigenstates diverge. The weak eigenstates (d' , s' , b') of down, strange and bottom quarks which are associated to the weak interaction are related to the mass eigenstates (d , s , b) through the Cabibbo-Kobayashi-Maskawa (CKM) matrix V_{CKM} [1, 2].

$$\begin{bmatrix} d' \\ s' \\ b' \end{bmatrix} = V_{\text{CKM}} \begin{bmatrix} d \\ s \\ b \end{bmatrix} = \begin{bmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{bmatrix} \begin{bmatrix} d \\ s \\ b \end{bmatrix} \quad (0.1)$$

Utilising the elements of the CKM matrix it is possible to calculate the probability that a quark type i will transition into type j with $|V_{ij}|^2$.

For a unitary 3×3 matrix the CKM can be reduced from nine parameters to four: three mixing angles (θ_{12} , θ_{13} , θ_{23}) which dictate the couplings between each generation of quarks and one phase (δ) which introduces CP violation [2]. A number of different parametrisations are possible but the standard choice is [3],

$$V_{\text{CKM}} = \begin{bmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{bmatrix} \quad (0.2)$$

where $c_{ij} = \cos \theta_{ij}$ and $s_{ij} = \sin \theta_{ij}$. The mixing angles θ_{ij} can be selected to satisfy $0 \leq \theta_{ij} \leq \frac{\pi}{2}$ such that $c_{ij}, s_{ij} \geq 0$.

Through experiments it is known that there is hierarchical structure to the components of the CKM matrix,

$$s_{12} \ll s_{23} \ll s_{13} \quad (0.3)$$

and it is useful to further parametrise through the Wolfenstein parametrisation [4] (Equation 0.5). For this parametrisation the following definitions are used,

$$\begin{aligned} s_{12} &= \lambda \\ s_{23} &= A\lambda^2 \\ s_{13} &= A\lambda^3(\rho + i\eta) \end{aligned} \quad (0.4)$$

resulting in,

$$V_{\text{CKM}} = \begin{bmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{bmatrix} + O(\lambda^4). \quad (0.5)$$

Reducing the CKM to $O(\lambda^3)$ results in the loss of unity. However, unity can be restored by including $O(\lambda^5)$ [5] correction to the bottom left component of Equation 0.5,

$$V_{td} = A\lambda^3(1 - \bar{\rho} - i\bar{\eta}) \quad (0.6)$$

where

$$\begin{aligned} \bar{\rho} &= \rho(1 - \frac{\lambda^2}{2}) \\ \bar{\eta} &= \eta(1 - \frac{\lambda^2}{2}) \end{aligned} \quad (0.7)$$

The unitary nature of the CKM matrix imposes conditions on the row and columns

$$\begin{aligned} \sum_i V_{ij} V_{ik}^* &= \delta_{jk} \\ \sum_j V_{ij} V_{kj}^* &= \delta_{ik} \end{aligned} \quad (0.8)$$

where δ_{jk} denotes the Kronecker delta function, which equals 1 when $j = k$ and 0 otherwise. This results in six vanishing combinations of the CKM matrix which can be represented as triangles, with the most common being

$$V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0 \quad (0.9)$$

and can be modified with the Wolfenstein parametrisation (Equation 0.10) which is depicted in Figure 2.

$$1 + \frac{V_{td}V_{tb}^*}{V_{cd}V_{cb}^*} = -\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*} = \bar{\rho} + i\bar{\eta} \quad (0.10)$$

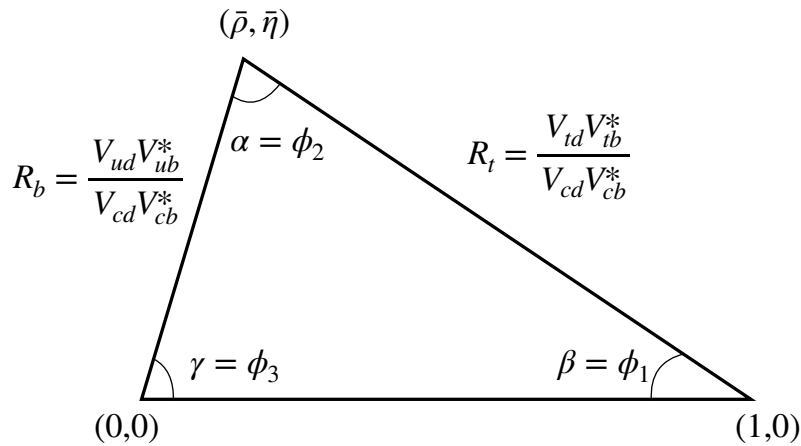


Figure 2: Sketch of the most common unitary triangle $V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0$.

The CKM matrix components are fundamental parameters of the SM. Through a large number of experiments the elements of the CKM matrix can be measured and are combined to form a global fit [6]:

$$\begin{bmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{bmatrix} = \begin{bmatrix} 0.97436^{+0.00014}_{-0.00019} & 0.22498^{+0.00081}_{-0.00062} & 0.00373^{+0.00019}_{-0.00015} \\ 0.22484^{+0.00081}_{-0.00061} & 0.97351^{+0.00016}_{-0.00019} & 0.04160^{+0.00066}_{-0.00121} \\ 0.00857^{+0.00018}_{-0.00037} & 0.04088^{+0.00066}_{-0.00119} & 0.999125^{+0.000052}_{-0.000027} \end{bmatrix} \quad (0.11)$$

A feature of the CKM matrix is that the diagonal components which describe the transitions within the quark generations dominate, compared to the off-diagonal elements which are suppressed for transitions between one generation and doubly suppressed for transitions between two generations. This suppression between two generations produces rare decays, such as $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ described in Section .

Through measurement of the CKM matrix components it is possible to constrain the unitary triangle (Figure 2) shown in Figure

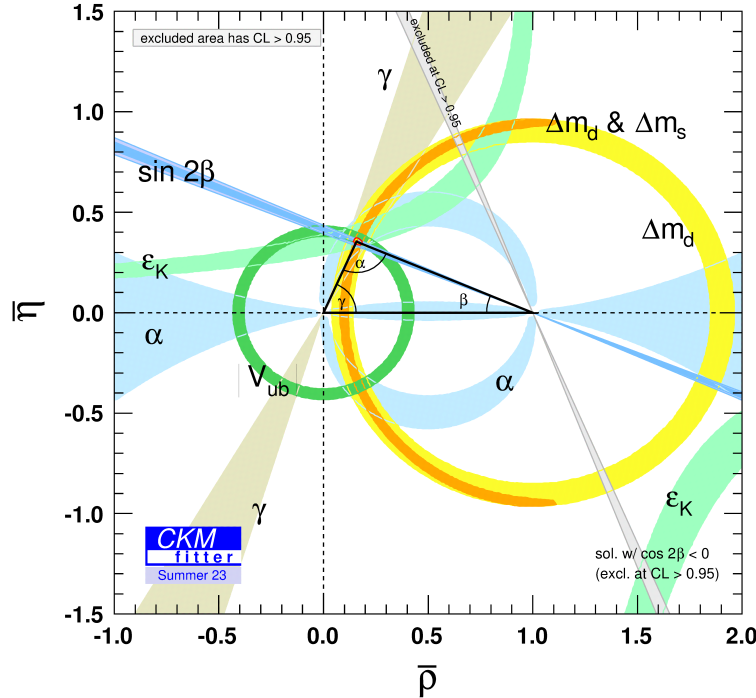


Figure 3: Constraints for the most common unitary triangle in the $(\bar{\rho}, \bar{\eta})$ plane [6].

Including the four CKM components discussed above there are nineteen free parameters of the SM:

- Four CKM parameters discussed above.
- Six Quark masses (Up, down, charm, strange, top and bottom).
- Three Lepton masses (Electron, muon and tau).
- Three Gauge Couplings (U(1), SU(2) and SU(3))
- Two Higgs parameters (Vacuum expectation value and mass).
- Quantum chromodynamics (QCD) vacuum angle.

Each one of these parameters needs to be measured through experiment to the highest precision so that any discrepancies can be resolved. Any deviations from theoretical understanding could be hint of physics beyond the Standard Model (BSM).

0.2 Kaon Physics

The first possible observation a heavy meson was in 1943 by Leprince-Ringuet and L'heritier [7, 8] and predated even the discovery of the pion in 1947 [9]. Utilising a cloud chamber to measure cosmic rays they found the first evidence of a particle with a mass of 990 times the mass of the electron (0.511 MeV [10]). However, it was possible that the upper limit on the measured mass could reach the proton mass so they were not credited with the discovery. In 1947 while similarly studying cosmic rays with cloud chambers G. D. Rochester and C. C. Butler published a result which included two pictures showing forked tracks [11]. One of the tracks proved to show the decay of a neutral particle into two charged daughter particles, which turned out to be decay of the neutral kaon to two pions. The other track showed the decay of a charged particle into charged and neutral daughter particles. This result was later confirmed in 1950 by a group in Cal-Tech who observed multiple instances of these decays [12]. However, between these publications a group in Bristol which included physicist Rosemary Brown, used emulsion chambers to observe the decay of a charged particle into three charged particles [13], known then as the tau decay. This resulted in what was known as the tau-theta problem, where the seemingly the same particle was decaying into two different final states with different parities:

Theta to two pions (+1) and Tau to three pions (-1). This eventually led to the discovery that parity is not conserved in weak interactions [14].

In the early 1950s these particles would be described as K-mesons, for which there are four types (K^+ , K^- , K^0 and $\bar{K}^0$). Experiments continued to measure the characteristics of the K mesons and found features such as long lifetimes (10^{-10} seconds) when compared to the fact their production was much faster [15]. These deviations from the other mesons, mainly the pion suggested the existence of a new quantum number named strangeness [16, 17]. The strange quantum number applies to all strongly interacting particles and is conserved in strong interactions but violated in weak interactions.

Into the latter 1950s and early 1960s the neutral kaon became the center of number of studies. Despite the fact that parity is violated in weak interactions it was still believed that the combination of charge-parity (CP) would be conserved. Through experiment the two weak eigenstates of the neutral kaon showed to have different lifetimes, which resulted in their names K_L and K_S [18]. Each of these eigenstates had different CP values with the K_S with $CP = +1$ decaying to two pions and the K_L with $CP = -1$ observed decaying to three pions. However, in 1964 it was observed that the K_L decayed to two pions [19] suggesting that CP was indeed violated in weak decays. This decay is an example of indirect CP violation as it occurs through neutral kaon mixing. The first evidence of direct CP violation, where the violation occurs within the decay was provided by the NA31 experiment in 1993 [20]. The process was eventually discovered by KTeV and NA48 collaborations in the latter 1990s [21, 22].

Kaon Type	Quark Content	Mass (MeV/c ²)	Lifetime (s)
K^+	$u\bar{s}$	493.677 ± 0.013	$(1.238 \pm 0.002) \times 10^{-8}$
K^-	$\bar{u}s$	493.677 ± 0.013	$(1.238 \pm 0.002) \times 10^{-8}$
K^0	$d\bar{s}$	497.611 ± 0.013	-
$\bar{K}^0$	$\bar{d}s$	497.611 ± 0.013	-
K_L^0	Mixed state of K^0 and $\bar{K}^0$	497.611 ± 0.013	$(5.116 \pm 0.021) \times 10^{-8}$
K_S^0	Mixed state of K^0 and $\bar{K}^0$	497.611 ± 0.013	$(0.8954 \pm 0.0004) \times 10^{-10}$

Table 1: Properties of charged and neutral kaons [10].

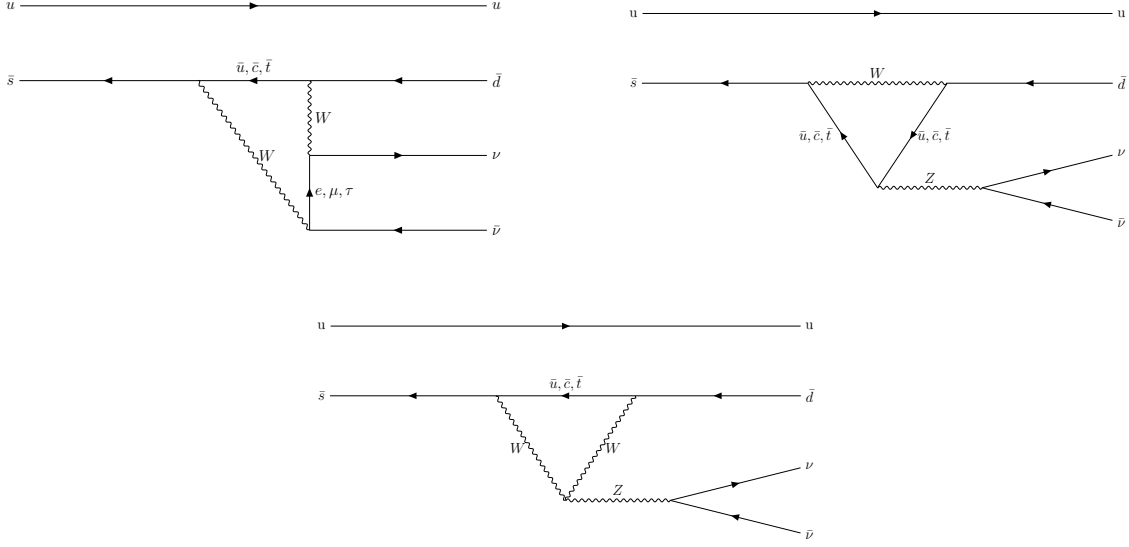
Kaon physics continued into the 21st century with the study of positively charged kaon decays which were conducted at NA62 at CERN. A description of the NA62 experiment is shown in Section ???. The positively charged kaon consists of an up quark and an anti-strange quark, with the other kaon types described in Table 1. The six most common decay modes of the positively charged kaon along with their shorthand notation is shown in Table 2.

Decay Mode	Shorthand Notation	Branching Ratio [%]
$K^+ \rightarrow \mu^+ \nu_\mu$	$K_{\mu 2}$	63.56 ± 0.11
$K^+ \rightarrow \pi^+ \pi^0$	$K_{2\pi}$	20.67 ± 0.08
$K^+ \rightarrow \pi^+ \pi^+ \pi^-$	$K_{3\pi}$	5.583 ± 0.024
$K^+ \rightarrow \pi^0 e^+ \nu_e$	K_{e3}	5.07 ± 0.04
$K^+ \rightarrow \pi^0 \mu^+ \nu_\mu$	$K_{\mu 3}$	3.352 ± 0.034

Table 2: Main decay modes of the positively charged kaon (K^+) with their shorthand notation and branching ratios [10].

As well as the decays found in Table 2 there are a number of extremely rare modes. NA62 main physics goal is to measure the Branching Fraction (BF) of the ultra-rare $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ which has a $O(10^{-11})$ and is described in more detail in Section 0.2.1.

The future kaon physics is planned to investigate the neutral kaon modes again. This was proposed through the HIKE proposal which is detailed in Section ??? which set out in multiple phases to measure the BF of $K_L^0 \rightarrow \pi^0 l^+ l^-$ (described in Section 0.2.2) and eventually $K_L^0 \rightarrow \pi^0 \nu \bar{\nu}$ (described in Section 0.2.1).

0.2.1 $K \rightarrow \pi \nu \bar{\nu}$ Figure 4: Feynman diagrams for the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay.

The $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decays through Flavour Changing Neutral Current (FCNC) which are extremely suppressed in the SM due to CKM and GIM suppression [23]. The decay process is described by the box and penguin Feynman diagrams shown in Figure 4. The decay is the transformation of a anti-strange quark into an anti-down quark and an neutrino pair, leaving the up quark unchanged. The branching ratio of $K_{\pi \nu \bar{\nu}}$ is dominated by the Z^0 penguin diagrams and after summing over the three neutrino flavours can be described by [24],

$$\mathcal{B}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = \kappa_+ (1 + \Delta_{EM}) \left[\left(\frac{\Im(\lambda_t)}{\lambda^5} X(x_t) \right)^2 + \left(\frac{\Re(\lambda_c)}{\lambda} P_c(X) + \frac{\Re(\lambda_t)}{\lambda^5} X(x_t) \right)^2 \right] \quad (0.12)$$

with

$$\kappa_+ = (5.173 \pm 0.025) \times 10^{-11} \left(\frac{\lambda}{0.225} \right)^8, \Delta_{EM} = -0.003 \quad (0.13)$$

where $x_t = m_t^2/m_W^2$ which is the ratio of the mass of the top quark and the mass of the W boson, λ originates from the Wolfenstein parametrisation shown above and $\lambda_i = V_{is}^* V_{id}$ where V_{ij} are CKM matrix elements. Δ_{EM} takes into account the correction from radiative photon exchanges and κ_+ condenses the relevant hadronic matrix elements. The terms $X(x_t)$ and $P_c(X)$ are the loop functions which describe

contributions from the top and charm quark. The $X(x_t)$ can be described by the equation [25, 24],

$$X(x_t) = X(0) + \frac{\alpha_s}{4\pi} X_t^{(1)} + \frac{\alpha}{4\pi} X_t^{(EW)} = 1.481 \pm 0.005 \pm 0.008. \quad (0.14)$$

It is possible to represent λ_t and λ_c from Equation 0.12 as [26],

$$\Re(\lambda_t) = -(1 - \frac{\lambda}{2}) A^2 \lambda^5 (1 - \bar{\rho}) \quad (0.15)$$

$$\Im(\lambda_t) = A^2 \lambda^5 \eta \quad (0.16)$$

$$\Re(\lambda_c) = -\lambda(1 - \frac{\lambda}{2}) \quad (0.17)$$

and can be used to parametrise the $K_{\pi\nu\bar{\nu}}$ branching fraction (0.12) with the CKM Wolfenstein parameters as [27]

$$\mathcal{B}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = \kappa_+ A^4 X(x_t)^2 \frac{1}{\sigma} ((\sigma^2 \bar{\eta})^2 + (\rho_0 - \bar{\rho})^2) \quad (0.18)$$

where $\sigma = (1/(1 - \frac{\lambda^2}{2}))^2$ and $\rho_0 = 1 + \frac{P_0(X)}{A^2 X(x_t)^2}$. Equation 0.18 represents an ellipse in the unitary triangle plane $(\bar{\rho}, \bar{\eta})$ with a centre of $(\rho_0, 0)$, a semi-major axis equal to $\sqrt{\frac{\mathcal{B}\sigma}{\kappa_+ A^4 X(x_t)^2}}$ and a semi-minor axis of $\sqrt{\frac{\mathcal{B}}{\sigma \kappa_+ A^4 X(x_t)^2}}$. The ellipse constrains the parameters $(\bar{\rho}, \bar{\eta})$ and is depicted in Figures 5 and 6.

The $K_L \rightarrow \pi^0 \nu \bar{\nu}$ ($K_{\pi^0 \nu \bar{\nu}}$) decays through similar feynman diagrams shown in Figure 4 but with a down quark on top which does not change during the decay compared to an up quark for the charged mode. The $K_{\pi^0 \nu \bar{\nu}}$ decay is almost entirely CP violating and is dominated by top loop contributions [28], resulting in the branching fraction depending on imaginary part of the top loop from Equation 0.12 which is CP violating,

$$\mathcal{B}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = \kappa_L \left(\frac{\Im(\lambda_t)}{\lambda^5} X(x_t) \right)^2 \quad (0.19)$$

with

$$\kappa_L = (2.231 \pm 0.013) \times 10^{-10} \left(\frac{\lambda}{0.225} \right)^8. \quad (0.20)$$

It is also possible to parametrise the equation with the Wolfenstein parameters similar to Equation 0.18 as [27],

$$\mathcal{B}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = \kappa_L \eta^2 A^4 X(x_t)^2. \quad (0.21)$$

The decay can be used to further constrain the parameter $\bar{\eta}$ and is shown in Figures 5 and 6.

Inputting experimental values into the equations for the branching fractions produces the SM predictions [29],

$$\mathcal{B}(K^+ \rightarrow \pi^+ \nu \bar{\nu}) = (8.60 \pm 0.42) \times 10^{-11} \quad (0.22)$$

and

$$\mathcal{B}(K_L \rightarrow \pi^0 \nu \bar{\nu}) = (2.94 \pm 0.15) \times 10^{-11}. \quad (0.23)$$

The first observation and measurement of the branching fraction for the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay has been obtained by NA62 in 2024 and is described in Section ???. The current best experimental result for the branching fraction for the $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decay is an upper limit obtained by the KOTO experiment [30],

$$\mathcal{B}(K_L \rightarrow \pi^0 \nu \bar{\nu}) < 300 \times 10^{-11} (90\% \text{CL}). \quad (0.24)$$

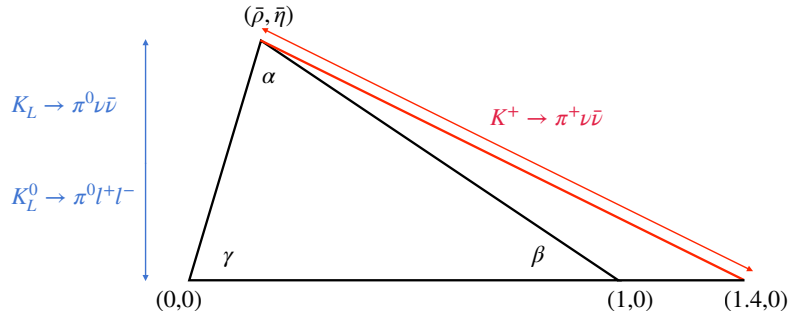


Figure 5: Diagram of the unitary triangle and its relationship with rare kaon decays.

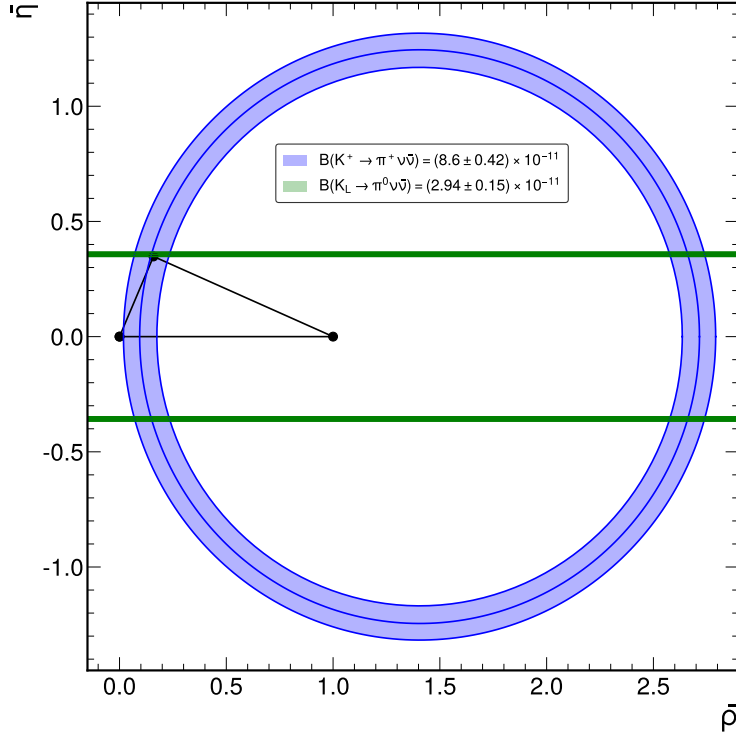


Figure 6: Constraints on the unitary triangle from the SM predictions for the branching fractions of $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ and $K_L \rightarrow \pi^0 \nu \bar{\nu}$ decays [31].

0.2.2 $K_L^0 \rightarrow \pi^0 l^+ l^-$

The decay $K_L^0 \rightarrow \pi^0 l^+ l^-$ is a FCNC transition of a strange quark into a down quark and a lepton-lepton pair and are theoretically clean with contributions through three processes: direct CP violating, indirect CP violating and CP conserving components.

The SM branching fraction for the $K_L^0 \rightarrow \pi^0 l^+ l^-$ decay can be described as [32, 33],

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 l^+ l^-) = \left(C_{mix}^l \pm C_{int}^l \left(\frac{\Im(\lambda_t)}{10^{-4}} \right) + C_{dir}^l \left(\frac{\Im(\lambda_t)}{10^{-4}} \right) + C_{CPC}^l \right) \times 10^{-12} \quad (0.25)$$

where C_{mix}^l represents the contributions due to $K_S - K_L$ mixing, C_{int}^l the interference of the indirect and direct CP violating decays, C_{dir}^l the contribution of direct CP violating and C_{CPC}^l is the long distance CP conserving contribution. The $|a_S|$ term depicts the decay form factor which has been calculated from results at NA48/2 from the measurement of the branching fraction of $K_S \rightarrow \pi^0 l^+ l^-$ as $|a_S| = 1.2 \pm 0.2$ [34, 35]. This results in the branching fractions for the $K_L^0 \rightarrow \pi^0 e^+ e^-$ and $K_L^0 \rightarrow$

$\pi^0 \mu^+ \mu^-$ decays,

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 e^+ e^-) = \left(15.7 |a_S|^2 \pm 6.2 |a_S| \left(\frac{\Im(\lambda_t)}{10^{-4}} \right) + 2.4 \left(\frac{\Im(\lambda_t)}{10^{-4}} \right) \right) \times 10^{-12} \quad (0.26)$$

and

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 \mu^+ \mu^-) = \left(3.7 |a_S|^2 \pm 1.6 |a_S| \left(\frac{\Im(\lambda_t)}{10^{-4}} \right) + 1.0 \left(\frac{\Im(\lambda_t)}{10^{-4}} \right) + 5.2 \right) \times 10^{-12}. \quad (0.27)$$

Equations 0.26 and 0.27 show the main difference between the two decay modes, that the CP conserving contribution due to the two-photon intermediate state only occurs in the muon decay as it is negligible in the helicity suppressed electron mode.

Utilising the above equations, set out in [36] the expected SM branching fractions can be described as,

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 e^+ e^-) = 3.54_{-0.85}^{+0.98} (1.56_{-0.49}^{+0.62}) \times 10^{-11} \quad (0.28)$$

and

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 \mu^+ \mu^-) = 1.41_{-0.26}^{+0.28} (0.95_{-0.21}^{+0.22}) \times 10^{-11}. \quad (0.29)$$

The current best results experimentally for the branching fraction of $K_L^0 \rightarrow \pi^0 l^+ l^-$ decays are the upper limits obtained by the KTeV experiment [37, 38]

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 e^+ e^-) < 28 \times 10^{-11} (90\% \text{CL}) \quad (0.30)$$

and

$$\mathcal{B}(K_L^0 \rightarrow \pi^0 \mu^+ \mu^-) < 38 \times 10^{-11} (90\% \text{CL}). \quad (0.31)$$

0.3 Dark Photon

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